

Prediction of Chronic Disease Risk Using Nutrition, Lifestyle & Clinical Data from The Irish Longitudinal Study on Ageing (TILDA)



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Submitted to the University of Limerick for the degree of

Artificial Intelligence (MSc) 2026

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Abstract

Falls, frailty, and chronic conditions such as cardiovascular disease, diabetes, and obesity remain leading causes of morbidity and mortality in older adults. Yet, existing predictive models often rely narrowly on physical health indicators, overlooking nutrition and lifestyle factors that may enhance early detection of risk. This thesis develops and evaluates machine learning models using data from The Irish Longitudinal Study on Ageing (TILDA) to predict key health outcomes in a cohort of older adults. The models integrate physical, mental, lifestyle, and socio-demographic predictors, with particular emphasis on under-explored nutritional and behavioural variables highlighted in prior research. Multiple approaches are compared, including logistic regression, random forests, and gradient boosting, with performance assessed via cross-validation and feature importance analysis. By identifying high-impact, modifiable predictors, this work aims to advance early risk stratification, guide targeted prevention strategies, and inform public health policy to support healthy ageing.

Declaration

I Abderahman Haouit declare that I have produced this thesis without the prohibited assistance of third parties and without making use of aids other than those specified; notions taken over directly or indirectly from other sources have been identified as such. This thesis has not previously been presented in identical or similar form to any other Irish or foreign examination board.

The thesis work was conducted from 2025 to 2026 under the supervision of Dr. Alison O'Connor at University of Limerick.

Limerick, 2026

Acknowledgements

I would like to acknowledge my supervisor, Dr. Alison O'Connor, for her guidance, support, and for providing a well-organised environment throughout the implementation of this project.

I would also like to thank the UL Science and Engineering Research Ethics Committee for approving the ethical aspects of this research. Special thanks to the Irish Social Science Data Archive (ISSDA) team for facilitating access to the TILDA dataset, collected by Trinity College Dublin.

Your dedication goes here. Open the dedication.tex file found at:
0_frontmatter/dedication.tex

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List of Abbreviations

AI	Artificial Intelligence
AUC	Area Under the ROC Curve
DNN	Deep Neural Network
ELSA	The English Longitudinal Study of Aging
EU	European Union
GBM	Gradient Boosting Machine
GDPR	General Data Protection Regulation
HRS	The Health and Retirement Study, (a longitudinal study of ageing in the United States)
IDS	The Intellectual Disability Supplement
ISSDA	The Irish Social Science Data Archive
LDA	Linear Discriminant Analysis
ML	Machine Learning
MPOD	Macular pigment optical density
RF	Random Forest
SHAP	Shapley Additive Explanations
SHARE	The Survey of Health, Aging and Retirement in Europe
TILDA	The Irish Longitudinal Study on Ageing
WHO	World Health Organization

List of Abbreviations

Chapter 1

Introduction

1.1 Background and Context

Chronic conditions—including cardiovascular disease, diabetes, and frailty—are leading contributors to morbidity and mortality in ageing populations [1, 2]. These conditions are associated with declines in functional capacity (e.g., gait speed, grip strength [3]), increased healthcare utilisation [4], and reduced quality of life [5].

The financial and societal burden is substantial. In Europe, poor diet is a major risk factor for non-communicable diseases, contributing billions in healthcare costs annually [6]. In Ireland, malnutrition alone is estimated to cost over €1.4 billion per year [7]. Recognising this, international and national policies—including the World Health Organization (WHO) Global Action Plan on Noncommunicable Diseases and Ireland’s *Healthy Ireland* framework—highlights nutrition and lifestyle interventions as central to healthcare strategy.

In parallel, (Artificial Intelligence (AI)) and (Machine Learning (ML)) are increasingly applied in healthcare to support risk prediction and personalised intervention. However, European regulations such as the General Data Protection Regulation (GDPR) and the forthcoming European Union (EU) AI Act impose strict requirements for transparency, explainability, and governance. This regulatory context underscores the importance of interpretable and ethically robust ML approaches.

1. INTRODUCTION

Traditional risk prediction models have largely relied on clinical biomarkers such as blood pressure and lipid profiles. Yet, growing evidence shows that broader predictors—including nutrition [8], physical activity [9], mental health [10], and social determinants [11]—enhance predictive performance.

(The Irish Longitudinal Study on Ageing (TILDA))¹ provides a nationally representative cohort of adults aged 50+ in Ireland, surveyed across multiple domains in repeated waves (waves 1–5). The raw wave datasets capture the full richness of TILDA-specific measures relevant to Irish health, policy, and society. In parallel, the Harmonized TILDA dataset restructures selected variables to align with the RAND (The Health and Retirement Study, (a longitudinal study of ageing in the United States) (HRS)) framework, enabling cross-national comparison with equivalent studies such as HRS (US), The English Longitudinal Study of Aging (ELSA) (UK), The Survey of Health, Aging and Retirement in Europe (SHARE) (continental Europe). This harmonisation enhances international comparability, though at the cost of excluding some study-specific variables. An additional dataset, the The Intellectual Disability Supplement (IDS), extends coverage to individuals with disabilities, improving inclusivity and reducing sampling bias. Together, these resources offer both detailed national data and a platform for internationally comparable, generalisable analyses.

Despite these advantages, existing ML research on chronic disease prediction faces persistent limitations: over-reliance on clinical predictors, limited incorporation of lifestyle and social variables, predominantly cross-sectional study designs that fail to capture trajectories, and a lack of interpretability consistent with emerging AI governance standards.

This thesis addresses these gaps by integrating TILDA’s longitudinal, multidomain data with interpretable ML methods, demonstrating how nutrition, physical activity, and mental health can be combined with traditional clinical markers to improve prediction of ageing-related outcomes.

¹TILDA data are pseudonymised and accessed under licence via The Irish Social Science Data Archive (ISSDA). Use adheres to UL S&E ethics approval and data governance requirements.

1.2 Research Problem and Motivation

Despite increasing policy interest in lifestyle and nutrition and advances in ML methods, approaches to chronic disease prediction in older adults remain constrained in several ways.

1.2.1 Limited Integration of Multidimensional Data

Most models prioritise clinical variables while underutilising modifiable lifestyle and nutrition factors [8, 12].

1.2.2 Underuse of Longitudinal Dynamics

Few studies exploit repeated measures to capture trajectories of risk, such as frailty progression [13].

1.2.3 Lack of Explainability

Many ML models are not interpretable, limiting clinical adoption and failing to meet emerging regulatory requirements for trustworthy AI in healthcare [14].

Addressing these challenges is essential to improve predictive accuracy, support evidence-based interventions, and ensure alignment with evolving policy and regulatory priorities.

1.3 Research Aims and Objectives

This thesis develops and validates ML models that integrate nutrition, lifestyle, and mental health features with clinical data to improve prediction of chronic disease and ageing-related outcomes in a representative cohort of older adults.

- Review existing applications of ML for chronic disease prediction in ageing populations, with emphasis on gaps in lifestyle and nutrition integration.
- Preprocess and harmonise TILDA waves 1-5, and engineer multidomain features (nutrition, activity, mobility, mental health, clinical indicators).

1. INTRODUCTION

- Develop and compare baseline statistical models (e.g., logistic regression) with advanced ML methods (Random Forest (RF), Gradient Boosting Machine (GBM)).
- Evaluate models using discrimination (Area Under the ROC Curve (AUC)), calibration, and error metrics (accuracy, precision, recall, F1-score), perform ablation analyses to assess the incremental value of lifestyle and nutrition predictors.
- Identify actionable predictors (e.g., vitamin D status, gait speed reserve) and highlight modifiable risk factors for intervention.
- Document ethical, governance, and reproducibility practices for handling longitudinal pseudonymised health data.

1.4 Contributions of the Research

This research makes contributions at three levels:

- **Empirical:** Demonstrates how nutrition, lifestyle, and mental health features enhance prediction when combined with established clinical baselines.
- **Methodological:** Provides a reproducible pipeline for harmonising longitudinal TILDA data and integrating multidomain features into ML workflows.
- **Clinical:** Identifies modifiable predictors with relevance for targeted interventions and policy to promote healthy ageing.

1.5 Thesis Outline

- *Chapter Two:* Literature review on ML prediction of chronic disease in ageing populations, with emphasis on lifestyle and nutrition predictors and methodological considerations.

- *Chapter Three:* TILDA study design, data preprocessing, harmonisation, and feature engineering.(Flow Charts..)
- *Chapter Four:* Modelling framework, including baseline and advanced ML, hyperparameter tuning, validation, and evaluation metrics.(Code snippets..)
- *Chapter Five:* Results, model comparison, feature importance, and ablation analyses.
- *Chapter Six:* Discussion of implications, limitations, ethical considerations, and future directions.

Supporting documents are included in the appendix:

- *Appendix A:* Variable codebook and wave harmonisation notes (including anonymisation actions, top-coding, and grouping rules relevant to the features used).
- *Appendix B:* Ethics.
- *Appendix C:* Modelling details, supplementary tables/figures, and reproducibility checklist (overview of data processing scripts).

A digital archive accompanies this dissertation (access request required):

- *Folder 1:* Preprocessing and feature-engineering scripts (documentation only, no raw TILDA data).
- *Folder 2:* Model training/evaluation code and configuration files to reproduce reported results.

1. INTRODUCTION

Chapter 2

Literature Review

2.1 Introduction

- Purpose of this chapter
- Brief overview on TILDA as a unique dataset (rich, longitudinal, multi-domai...)
- Scope: frailty, mental health, chronic disease, multimorbidity, nutrition, and ML models

2.2 Chronic Disease and Ageing

- The burden of chronic conditions in older adults
- Healthy ageing framework WHO
- Importance of multimorbidity, frailty, mental health

2.2.1 Frailty and Physical Function

- Gait speed, grip strength, functional mobility decline
- Predictors of frailty transitions ([15], [16], [17])

2. LITERATURE REVIEW

2.2.2 Mental Health in Older Adults

- Depression, anxiety, loneliness, social connectedness ([18], [19], [20])
- Links between physical activity and depression ([10], [9], [12])

2.2.3 Nutrition and Biological Ageing

- Vitamin D, lutein, zeaxanthin, Macular pigment optical density (MPOD) ([8], [21], lairdVitaminDeficiencyIreland2020, feeneyLowMacularPigment2013)
- Malnutrition and its predictors ([2])
- Biological vs chronological ageing ([22], [23])

2.3 Machine Learning in Ageing and Chronic Disease

- Overview: Why ML is valuable in ageing research [24]
- Transition from regression-based models → ensemble ML → deep learning(cite a example here)

2.3.1 Feature Sets in ML Studies

- Clinical, sociodemographic, lifestyle, cognition, psychosocial
- Over-reliance on physical health predictors vs underuse of nutrition/social variables

2.3.2 ML Models Used

- Logistic regression, Markov models, Linear Discriminant Analysis (LDA) (traditional)
- Ensembles RF, GBM, Deep Neural Network (DNN) – e.g., ([25]) diabetes prediction

- Unsupervised clustering for dementia ([26])
- Brain age prediction models ([27], [28])

2.3.3 Performance and Validation

- AUC, R^2 ranges across tasks (diabetes, gait speed, mortality, depression)
- Limitations: internal validation only, cross-sectional data, no generalisation

2.3.4 Explainability in ML for Ageing

- Shapley Additive Explanations (SHAP), TreeExplainer examples ([25], [29], [3])
- Challenges of black-box models in clinical practice

2.4 Gaps in the Literature

- Limited population diversity (mostly Irish, 50+)
- Over-reliance on physical predictors; neglect of nutrition, social, religious/spiritual, environmental variables
- Dominance of regression models; limited advanced ML such as neural networks, transformers
- Validation gaps – very few external validations, especially for ensemble and brain-age models
- Underexplored outcomes: pain, sleep, QoL
- Cross-sectional bias; weak time-series modelling
- Poor transparency on missing data handling
- Limited integration of nutrition biomarkers
- Lack of biological age prediction vs chronological

2. LITERATURE REVIEW

- Interpretability challenges in ML

2.5 Chapter Summary

- I need to recap the main findings from LR systematic mapping excel file
- Identify your contribution niche:
 - integrating nutrition + lifestyle + clinical predictors
 - ML + explainability
 - time-based prediction (longitudinal TILDA data)
- Lead into Methodology (Chapter 3)

Chapter 3

Data and Preprocessing

3.1 Insert a Figure

We refer to a figure in the text like this: (Figure 3.1).



Figure 3.1: Title of Figure - Description. [Source: (?)]

3. DATA AND PREPROCESSING

3.2 Creating a list

- (2000) item 1.
- (2004) item 2.
- (2010) item 3.
- (2013) item 4.

3.2.1 Creating a Table

Table 3.1: Table Title

	Resolution	Min	Max
Gyroscope	4.5mV / °/s	0.27 °/s	406 °/s
Accelerometer	600mV /g	0.002g	2g
Magnetometer	385mV/ gauss	0.317 gauss	6 gauss

3.3 Quote

An in-text quote is declared in the following way:

Recycling is the way forward! Recycling is the way forward! Recycling
is the way forward! Recycling is the way forward! Recycling is the
way forward! Recycling is the way forward! Recycling is the way
forward! Recycling is the way forward! Recycling is the way forward!
Recycling is the way forward! Recycling is the way forward! Recycling
is the way forward! Recycling is the way forward! Recycling is the
way forward! Recycling is the way forward! Recycling is the way
forward! Recycling is the way forward! Recycling is the way forward!
Recycling is the way forward! (CommonSense, p. 0)

Another way of doing it is:

*There are in our existence spots of time,
That with distinct pre-eminence retain
A renovating virtue, whence-depressed
By false opinion and contentious thought,
Or aught of heavier or more deadly weight,
In trivial occupations, and the round
Of ordinary intercourse-our minds
Are nourished and invisibly repaired;
A virtue, by which pleasure is enhanced,
That penetrates, enables us to mount,
When high, more high, and lifts us up when fallen.*

(? , verses 208-218)

3. DATA AND PREPROCESSING

Chapter 4

Methodology

4.1 Math

In-text Math

- A vector vector $\hat{A}_{ba}\{x_{ba}, y_{ba}, z_{ba}\}$
- Another vector $R^b\{\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}\}$ as:

A series of Equations:

$$\hat{t}_{1b} = \hat{A}_{ba} \tag{4.1}$$

$$\hat{t}_{2b} = \frac{\hat{A}_{ba} \times \hat{M}_{bm}}{|\hat{A}_{ba} \times \hat{M}_{bm}|} \tag{4.2}$$

$$\hat{t}_{3b} = \hat{t}_{1b} \times \hat{t}_{2b} \tag{4.3}$$

$$R^a = R^b(R^i)^T = [\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}][\hat{t}_{1i}, \hat{t}_{2i}, \hat{t}_{3i}]^T \tag{4.4}$$

where the symbol T denotes transposition.

$$\hat{t}_{2b} = \frac{\hat{A}_{ba} \times \hat{M}_{bm}}{|\hat{A}_{ba} \times \hat{M}_{bm}|} \tag{4.5}$$

4. METHODOLOGY

$$\begin{aligned}
&= \frac{(y_{ba}z_{bm} - y_{bm}z_{ba}, z_{ba}x_{bm} - z_{bm}x_{ba}, x_{ba}y_{bm} - x_{bm}y_{ba})}{|A_{ba}| |M_{bm}| \sqrt{1 - (\hat{A}_{ba} \cdot \hat{M}_{bm})^2}} \\
&= \frac{-0.2338, -0.0931, -0.0651}{0.2601} \\
&= (-0.8989, -0.3579, -0.2503)
\end{aligned}$$

which if normalised gives:

$$= (-0.8995, -0.3581, -0.2504)$$

A matrix

$$R^b = [\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}] = \begin{pmatrix} -0.2698 & -0.8995 & 0.3439 \\ 0.0035 & -0.3581 & -0.9337 \\ 0.9629 & -0.2504 & 0.0997 \end{pmatrix}$$

Chapter 5

Results

5.1 References

Open the file ‘references_myphd’ with BibTex. The file is located in ‘/7_references/’

The file contains a series of templates for the proper formatting of references according to the Harvard referencing style. Click on each reference in the text below to see how Latex has formatted the reference on the References section.

- **Book:** (? , p. 10) (page number required only if referring to an in-text quote)
- **Book Chapter:** (?)
- **Book Contribution:** (?)
- **MUSIC CD/DVD:** (?)
- **Journal Article:** (?)
- **Patent:** (?)
- **PhD/Master Dissertation:** (?)
- **Conference Paper:** (?)
- **Technical Report:** (?)

5. RESULTS

- **Unpublished Report:** (?)
- **Web Videos**(?) (. . . not the name of the person who uploaded the video though)
- **Web Paper:** (?)
- **Web Journal Article:** (?)
- **Webpage:** (?)
- **Downloaded Images or Sounds:** (?)

You can also use this kind of referencing if needed in the text:
Surname [?] said that . . .

Chapter 6

Conclusions and Future Directions

6.1 Summary

6.2 Conclusions

6.3 Contributions

6.4 Future Work

6. CONCLUSIONS AND FUTURE DIRECTIONS

Appendix A

Insert a figure

Appendix

Appendix B

Title

Type here you text as usual . . .

Appendix

Appendix C

Code for estimating attitude

AHRS_TRIAD.C

```
/*
 *
 * Copyright (c) 2013, Giuseppe Torre
 * All rights reserved.
 *
 * Redistribution and use in source and binary forms, with or without
 * modification, are permitted provided that the following conditions are met:
 *     * Redistributions of source code must retain the above copyright
 *       notice, this list of conditions and the following disclaimer.
 *     * Redistributions in binary form must reproduce the above copyright
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 * (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES;
 * LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND
 * ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT
 * (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS
 * SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
 */

#include "ext.h"
#include "ext.mess.h"
#include <string.h>
#include <stdio.h>
#include <math.h>
#define EPSILON 1e-6

//*****CALIBRATION STUFF*****//
/* INTRODUCE THE OFFSET VALUES IN THE FOLLOWING VARIABLES IN ADC values */
#define ACCELX.OFFSET 2043 //initial offset in accelerometer X
#define ACCELY.OFFSET 2053 // initial offset in accelerometer Y
#define ACCELZ.OFFSET 2264//initial offset in accelerometer Z

#define MagX.OFFSET 1917 //initial offset in Magnetometer X
```

Appendix

```
#define MagY_OFFSET 1813 // initial offset in Magnetometer Y
#define MagZ_OFFSET 1667 // initial offset in Magnetometer Z
/* WE SHOULD CALIBRATE THE FOLLOWING VALUES FOR EACH PARTICULAR IMU axis */
#define ACCELX_Resolution 536
#define ACCELY_Resolution 661
#define ACCELZ_Resolution 559
#define MagX_Resolution 186
#define MagY_Resolution 189
#define MagZ_Resolution 170
// ***** END CALIBRATION STUFF *****//

void *this_class; // Required. Global pointing to this class

typedef struct _triad // Data structure for this object
{
    t_object m_ob; // Must always be the first field; used by Max

    Atom m_args[9]; // we want our inlet to be receiving a list of 10 elements

    long m_value; //inlet

    void *m_R1; //these are all the outlets for the 3 X 3 Matrix
    void *m_R2; // R1 -> firts top left cell ..R2 middle cell of the first
    row ...and so on
    void *m_R3; //
    void *m_R4; // End Outlets
} t_triad;

void *triad_new(long value);

void triad_assist(t_triad *triad, void *b, long msg, long arg, char *
s);
void triad_free(t_triad *triad);

void triad_list(t_triad *x, Symbol *s, short argc, t_atom *argv);

void MatrixByMatrix(double *Result, double *MatrixLeft, double *
MatrixRight);

void Matrix2Quat(double *Quat, double *Matrix);
void Quat2Matrix(double *Matrix, double *Quat);
void inverseQuat(double *InvQuat, double *RegQuat);
void NormQuat(double *YesQuat, double *NotQuat);
void Slerp(double *NewQuat, double *OldQuat, double *CurrentQuat);
void NormVect(double *YesVect, double *NotVect);
double orientationMatrix[9];
double Result[9];
int i;
double InvorientationMatrix[9];

double temp[6];
double ref[6];
double vectAx[3];
double vectAy[3];
double vectAz[3];
double vectBx[3];
double vectBy[3];
double vectBz[3];
double MagnCrosProd_A;
double MagnCrosProd_B;
double accnorm, magnorm, earthnorm, VectAynorm, VectAznorm,
VectBynorm, VectBznorm;
```

```

        double m[9], n[9];
        double quat_e[4];
        double invquat_e[4];
        double mult;
        double quat_new[4];
        double quat_old[4];
        double ecs, accex_ADCnumber, y, accey_ADCnumber, z, accez_ADCnumber, mx,
            magnx_ADCnumber, my, magny_ADCnumber, mz, magnz_ADCnumber;

// SLERP Variables

        double trace, Suca;
        double tol[4], omega, sinom, cosom, scale0, scale1, tez,
            orientationMatrixA[9];

int main(void)
{
    // set up our class: create a class definition
    setup((t_messlist**) &this_class, (method)triad_new, (method)triad_free, (short)
        sizeof(t_triad), 0L, A_GIMME, 0);
    address((method)triad_list, "list", A_GIMME, 0);
    address((method)triad_assist, "assist", A_CANT, 0);
    finder_addclass("Maths", "triad");
    post(".... I 'm-TRIAD-Object !.... from -AHRS- Library ...", 0);
    return 0;
}

/* ----- triad_new ----- */

void *triad_new(long value)
{
    t_triad *triad;
    triad = (t_triad *)newobject(this_class);           // create the new instance and return
        a pointer to it
    triad->m_R4 = floatout(triad);
    triad->m_R3 = floatout(triad);
    triad->m_R2 = floatout(triad);
    triad->m_R1 = floatout(triad);

    return(triad);
}

etc.

etc.

```

Appendix

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